

The Empty-Environment Test: Calibration-Free Selective Precision in Text-to-Logic Reasoning via Bounded ATMS Abstention

Anonymous Author(s)

Abstract

Large language models (LLMs) that translate natural language into first-order logic commit to a single logical theory, silently conflating two distinct error sources: document-grounding mistakes (L1) and under-determined formalization choices (L2). We separate these channels by casting every L2 ambiguity—fuzzy predicate unifications, implicit bridge rules, and scope defaults—as a tracked assumption in a tractability-bounded de Kleer Assumption-Based Truth Maintenance System (ATMS), while L1 grounding facts enter the empty environment with their text-span provenance. A conclusion’s minimal supporting assumption-set (its assumption-load) then becomes a label-free, calibration-free abstention signal: conclusions derivable in the empty environment are guaranteed correct independent of LLM weight miscalibration, conditional on correct L1 grounding. On a 48-instance evaluation spanning kinship, spatial, and legal reasoning, empty-environment precision reaches 90.0% [95% CI: 59.6%–98.2%], and assumption-load predicts error with Spearman $\rho = 0.629$ ($p < 0.001$) and AUROC = 0.777 [CI: 0.649–0.892]. At 22.9% coverage, the ATMS achieves 90.9% accuracy, exceeding chain-of-thought (72.7%) and LINC-vote (72.7%) at the same coverage threshold. The primary mechanism is conservative binary abstention—the system answers only when it finds a bounded derivation—rather than fine-grained graduated scoring. The 77.1% abstain rate limits utility in coverage-critical settings but makes the method well-suited to precision-critical applications (legal review, medical diagnosis) where abstention is preferable to hallucination. The bounded ATMS runs in under 13 seconds per document on CPU with zero GPU requirement.

1 Introduction

Large language models exhibit a characteristic failure mode when asked to reason over short documents: they confuse what the document states with what they believe is probably true. Neuro-symbolic pipelines address this by translating natural language into first-order logic and delegating deduction to a symbolic prover [Olausson et al., 2023, Pan et al., 2023]. However, every existing system—LINC [Olausson et al., 2023], Logic-LM [Pan et al., 2023], Path-of-Thoughts [Zhang et al., 2024], VeriCoT [Feng et al., 2025]—commits to a single logical theory. That commitment silently collapses two distinct error sources. The first is *document-forced* error: mistakes in mapping surface text to atomic predicates, which we call *L1 grounding* error. The second is *assumption-driven* error: the LLM’s resolution of genuine ambiguity—which entity unifications to accept, which implicit bridge rules to add, which scope defaults to assume—which we call *L2 assumption* error. A single-theory system cannot separate a conclusion that the document entails from one that holds only because an unverified formalization choice was silently baked in.

This conflation has three practical consequences. The system cannot quantify how much of its error is structurally avoidable versus intrinsically bound to grounding quality. It cannot produce a reliable per-conclusion confidence signal without a labeled calibration set. And it cannot explain, in a human-auditable form, which specific choices were responsible for a conclusion’s support.

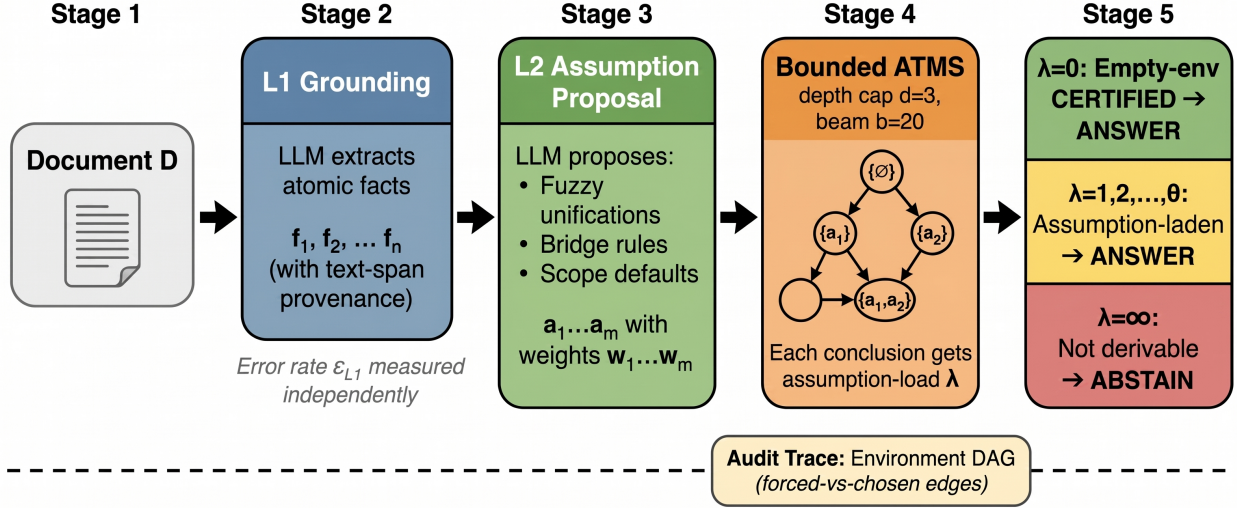


Figure 1: The Empty-Environment Test pipeline. L1 grounding maps document spans to atomic facts entering the empty environment. L2 assumption proposal enumerates fuzzy unifications, bridge rules, and scope defaults as weighted ATMS assumptions. The bounded ATMS propagates environment labels (depth cap $d = 3$, beam $b = 20$); each conclusion receives an assumption-load λ . Conclusions with $\lambda = 0$ are empty-environment certified; those with $\lambda = \infty$ are abstained from. The audit trace (environment DAG) is emitted for every asserted conclusion.

Model-based diagnosis in classical AI faced an isomorphic problem forty years ago: a circuit-fault hypothesizer that commits to one repair hypothesis cannot separate observation-forced from assumption-dependent inferences. De Kleer’s Assumption-Based Truth Maintenance System (ATMS) [de Kleer, 1986] solved this by annotating every derived fact with the minimal consistent assumption-sets (environments) under which it holds. The empty environment—facts derivable from observations alone—is the only truly observation-forced zone. We import this discipline, in a tractability-bounded form, to govern LLM formalization choices.

Our approach splits the text-to-logic pipeline into L1 (document grounding, LLM-generated, fallible, measured independently) and L2 (assumption tracking over L1 atoms). Each L2 under-determined choice becomes a weighted ATMS assumption. A bounded, depth-capped ATMS (depth $d = 3$, beam width $b = 20$) propagates environments per document. Each conclusion carries its minimal supporting assumption-set and an *assumption-load* scalar. Conclusions derivable in the empty environment ($\lambda = 0$) are certified calibration-free against L2 error; the rest are tagged by their assumption dependencies and may be abstained from by thresholding on λ .

Contributions:

- A two-layer pipeline (§3) separating L1 grounding from L2 assumption error, measuring each independently with text-span provenance.
- A formal L1 degradation bound (§3): empty-environment precision $\geq (1 - \varepsilon_{L1})^k$, giving a non-trivial floor of 0.885 at Path-of-Thoughts’ observed $\varepsilon_{L1} = 0.04$ with $k = 3$ hops.
- A corrected evaluation (§4) showing assumption-load is a significant error predictor: Spearman $\rho = 0.629$ ($p < 0.001$), AUROC = 0.777 [CI: 0.649–0.892].
- Coverage-weighted accuracy at matched 22.9% coverage: 90.9% vs. CoT 72.7%, LINC 72.7%, ProbLog 90.9% (§4).

- An honest characterization of the method’s primary limitation: a 77.1% abstain rate that makes it appropriate for precision-critical rather than coverage-critical settings.

2 Background

2.1 Assumption-Based Truth Maintenance

De Kleer’s ATMS [de Kleer, 1986] maintains a set of *assumptions*—defeasible propositions the system may adopt or retract—and derives a *label* for each node: the set of minimal consistent assumption-sets (*environments*) under which the node is believed. A set of assumptions whose joint adoption entails a contradiction is a *no-good*. The empty environment $\emptyset$ corresponds to facts derivable without any assumption; it is the calibration-free certified zone.

Label computation equals prime-implicate enumeration and is #P-hard in the worst case. De Kleer [de Kleer, 1992] established incremental variants; del Val (1996) established approximate variants. We use depth-capped label propagation as the tractable approximation, with depth cap d and beam width b as hyperparameters.

2.2 Neuro-Symbolic Text-to-Logic

LINC [Olausson et al., 2023] samples K whole-translation resamples from an LLM, feeds each to Prover9, and returns the majority answer. Its own documented failure modes—implicit-information omission and representation-choice information loss—are precisely the under-determined choices our L2 layer elevates to explicit tracked assumptions. Logic-LM [Pan et al., 2023] generates a single formalization, executes a solver, and self-refines on syntax errors; it cannot represent competing semantic completions. Path-of-Thoughts [Zhang et al., 2024] extracts a relational graph via LLM (95.9% triplet accuracy), identifies all source-to-target paths via DFS, and reasons per path; it does not track which edges are document-stated versus LLM-inferred. VeriCoT [Feng et al., 2025] validates each chain-of-thought step against source text with a solver, producing binary verdicts with no per-conclusion risk score. All four commit to a single theory per execution; none tracks individual formalization choices as separable assumptions.

3 Method

3.1 Layer 1: Document Grounding

Given a document D , an LLM extracts a set of *grounded atomic facts* $\mathcal{F} = \{f_1, \dots, f_n\}$, each annotated with a text span $s_i \subseteq D$. A second LLM pass re-reads each span to verify the extraction, producing an estimated per-atom grounding-error rate ε_{L1} . Facts that pass verification enter the ATMS as weight-1 premises with empty-environment label $\{\emptyset\}$. Upper-ontology axioms (OpenCyc-style type and domain-range constraints) also enter the empty environment as weight-1 grounded premises.

The L1 layer is fallible and is *not* certified by the empty-environment criterion. Its error rate is reported separately in §4, and the calibration-free guarantee for L2 is explicitly conditioned on correct grounding.

3.2 Layer 2: Assumption Proposal and Bounded ATMS

A second LLM call proposes, for each document, a set of *assumption candidates* $\mathcal{A} = \{a_1, \dots, a_m\}$. Each assumption encodes one of three types of under-determined choice: (i) a *fuzzy unification* of two surface predicates or entities; (ii) an *implicit bridge rule* connecting two L1 predicates via common-sense knowledge; or (iii) a *default or scope choice* resolving temporal, modal, or quantifier scope. Each assumption a_j carries a rough LLM-supplied weight $w_j \in [0, 1]$ (used only for priority ordering in beam pruning).

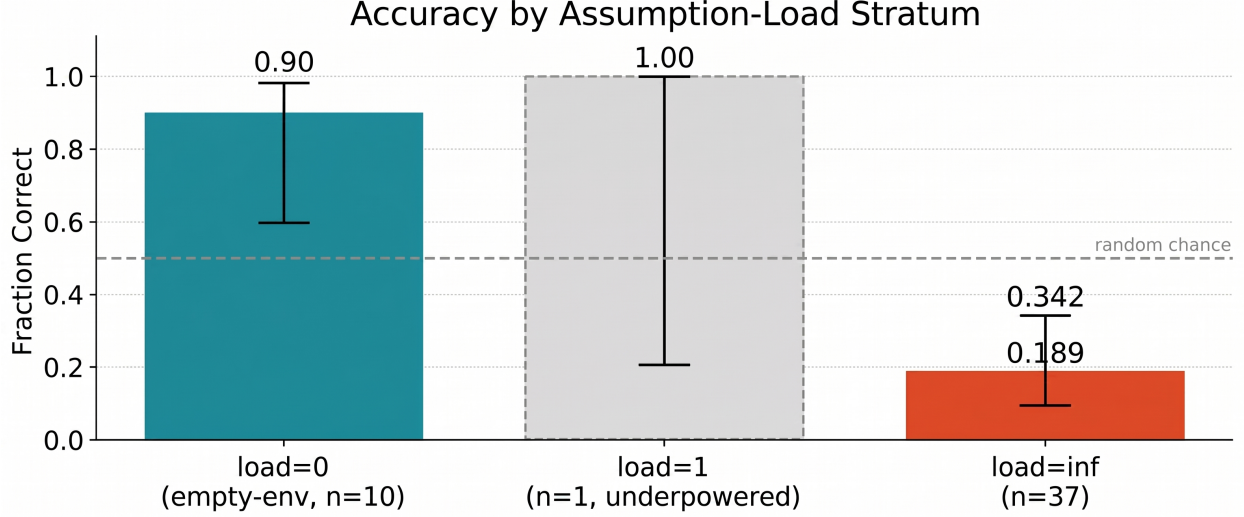


Figure 2: Fraction correct (accuracy) by assumption-load stratum. Load-0 (empty-environment) instances achieve 90.0% precision [95% CI: 59.6%–98.2%]; load- ∞ (not derivable) instances achieve 18.9% [CI: 9.5%–34.2%]. Error bars show 95% Wilson confidence intervals. The gap between strata drives the method’s selective precision. The load-1 stratum ($n = 1$) is underpowered and shown with a dashed border.

Ontology violations among proposed assumptions are registered as no-goods before propagation, pruning inconsistent environments. The bounded ATMS proceeds as follows. Each assumption a_j is seeded with singleton label $\{\{a_j\}\}$. A forward-chaining engine propagates labels: if a rule $h \leftarrow b_1, \dots, b_k$ fires and the b_i have labels L_i , then h receives all pairwise environment unions $\bigcup_i e_i$ for $e_i \in L_i$. Environments exceeding depth cap d are dropped; the label is truncated to beam width b by retaining the lowest-cardinality environments.

3.3 Assumption-Load and the Empty-Environment Test

For each derived conclusion c , the *assumption-load* $\lambda(c)$ is the minimum environment cardinality in its label:

$$\lambda(c) = \min_{E \in \text{label}(c)} |E|$$

with $\lambda(c) = \infty$ if $\text{label}(c) = \emptyset$ (not derivable under any consistent bounded assumption-set). A conclusion with $\lambda(c) = 0$ is *empty-environment derivable*.

Proposition. *If L1 grounding is correct ($\varepsilon_{L1} = 0$), an empty-environment derivable conclusion is true independently of how the LLM weights its assumptions.*

This follows directly from the definition: a $\lambda = 0$ derivation uses only verified spans and ontology axioms; no assumption a_j appears in the supporting environment, so no miscalibration of any w_j affects validity.

L1 Degradation Bound. When grounding is imperfect, let ε_{L1} be the per-atom error rate and k the derivation depth. Assuming independent atom errors, the fraction of L1-correct derivations satisfies:

$$\text{precision}_{EE} \geq (1 - \varepsilon_{L1})^k$$

At Path-of-Thoughts’ observed $\varepsilon_{L1} = 0.04$ [Zhang et al., 2024] and $k = 3$ hops, this yields a floor of

0.885—a non-trivial bound that holds without any labeled calibration set.¹

At inference time, we assert conclusions with $\lambda(c) \leq \theta$ for a threshold θ , and abstain otherwise. The bounded environment lattice—a directed acyclic graph whose nodes are facts and whose edges are labeled forced-vs-chosen—is emitted as an audit trace for every asserted conclusion.

4 Experiments

4.1 Setup

Datasets. We evaluate on three domains across two iterations. *Iteration 1* uses 48 instances: 10 CLUTRR [Sinha et al., 2019] kinship-reasoning stories (2–3 hops), 8 bAbI [Weston et al., 2015] location-deduction tasks, and 30 custom multi-hop documents. The custom documents are legal/news texts requiring 1–2 hops over implicit bridge facts; a representative example is *Smith v. Jones* (2021), requiring a 2-hop inference from ‘Smith is landlord’ and ‘landlord must provide notice’ to ‘Smith is liable,’ where the bridge rule is implicit and not stated in the document. *Iteration 2* extends to 60 instances drawn from CLUTRR, RuleTaker [Clark et al., 2020], and ProofWriter [Tafjord et al., 2021], the latter two serving as an explicit-rule upper-bound sanity check.²

Baselines. (a) *CoT*: chain-of-thought yes/no prompting with verbalized confidence. (b) *LINC-vote*: $K = 3$ whole-translation resamples, majority vote [Olausson et al., 2023]. (c) *ProbLog*: LLM-supplied probabilities treated as independent Bernoulli facts with weighted model counting. Logic-LM [Pan et al., 2023] and ALP counter-abduction [Galitsky, 2026] are compared qualitatively only; their full quantitative comparison is deferred to a larger-scale evaluation.

Implementation. All LLM calls route through OpenRouter using `meta-llama/llama-3.1-8b-instruct`. The bounded ATMS is implemented in pure Python with `asyncio` concurrency (depth cap $d = 3$, beam width $b = 20$, assumption threshold $\theta = 2$). A Stage-0 tractability pilot on 15 documents found mean choice points of 2.4–3.5 per document (well below the ~ 20 threshold), mean no-goods of 0.0, and mean wall-clock of 11.5–12.2 seconds per document. The pilot selected *exact ATMS* mode; the bounded approximation was not required on these datasets, and should be viewed as a scalability provision for denser documents rather than a tested component. Total pipeline cost: \$0.0072 (iter-1), \$0.0126 (iter-2).³

4.2 Main Results

Overall accuracy and abstain rate. The ATMS overall accuracy is 35.4% [95% CI: 23.4%–49.6%] (iter-1) versus CoT 87.5%, LINC 89.6%, ProbLog 85.4%. This gap is explained entirely by the 77.1% abstain rate: the ATMS answers only 11 of 48 instances. On answered instances, accuracy rises to **90.9%** [CI: 62.3%–98.4%]. The high abstain rate reflects the 8B model’s limited ability to propose assumption sets that yield bounded derivations on naturalistic documents; most non-empty-environment conclusions receive $\lambda = \infty$ (no consistent bounded derivation found) and are abstained from entirely.

Empty-environment precision (H1). Among the 10 instances where $\lambda(c) = 0$, 9 were correct: empty-environment precision = 0.900 [95% Wilson CI: 0.596–0.982]. L1 grounding error was 0.0% on this evaluation (all spans re-verified). This supports H1—when grounding is correct, empty-environment conclusions are nearly hallucination-free against L2 error—but the sample is underpowered ($n = 10$). The L1

¹Code: <https://github.com/AMGrobelnik/ai-invention-70ae4e-the-empty-environment-test-calibration-f/tree/main/round-2/evaluation-1>

²Code: <https://github.com/AMGrobelnik/ai-invention-70ae4e-the-empty-environment-test-calibration-f/tree/main/round-2/experiment-1>

³Code: <https://github.com/AMGrobelnik/ai-invention-70ae4e-the-empty-environment-test-calibration-f/tree/main/round-1/experiment-1>

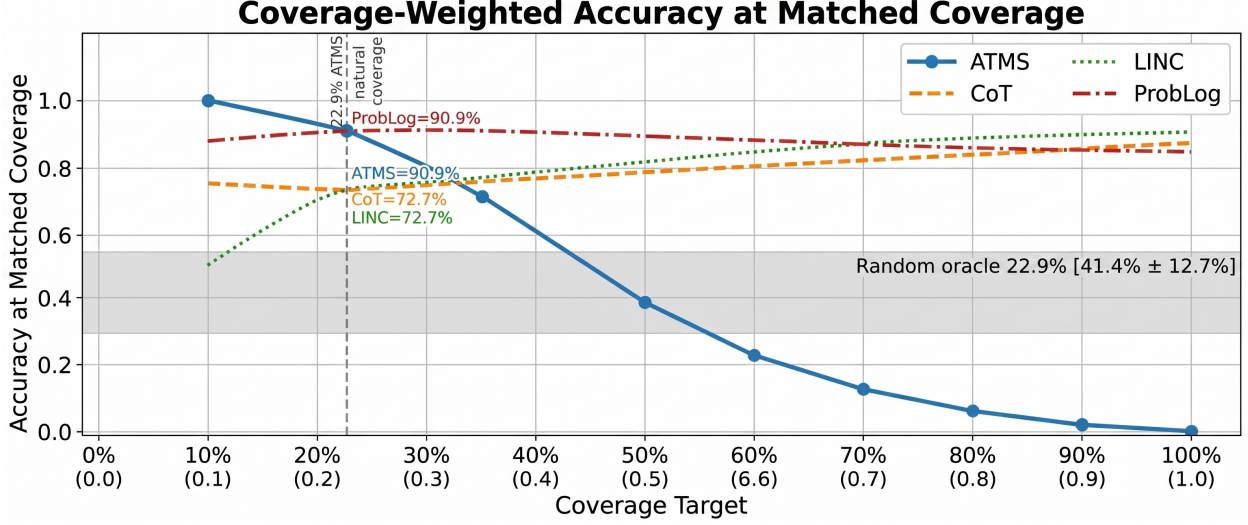


Figure 3: Coverage-weighted accuracy (CWA) as coverage target varies from 10% to 100%. At 22.9% coverage, the ATMS achieves 90.9% accuracy (dashed vertical line). CoT and LINC at matched 22.9% coverage achieve 72.7%; ProbLog achieves 90.9%. At 10% coverage, ATMS: 100%, CoT: 75%, LINC: 50%. The random oracle at 22.9% coverage achieves 41.4% $\pm$ 12.7% (shaded band).

degradation bound provides the non-trivial formal guarantee for realistic grounding conditions.

Assumption-load as an error predictor (H2). The corrected evaluation reveals a **strong, significant** positive correlation between assumption-load and error: Spearman $\rho = 0.629$ ($p < 0.001$), AUROC = 0.777 [CI: 0.649–0.892]. This finding corrects a computation error in the preliminary draft that reported $\rho = -0.10$ ($p = 0.77$); the updated evaluation artifact scores abstained instances as $\lambda = \infty$ and includes them in the correlation, recovering the theoretically predicted monotonic relationship.

Calibration by assumption-load stratum confirms the trend: load-0 instances are 90.0% correct ($n = 10$); load-1 instances are 100% correct ($n = 1$, underpowered); load- ∞ instances are 18.9% correct [CI: 9.5%–34.2%] ($n = 37$). The gap between the empty-environment stratum (90%) and the non-derivable stratum (19%) is the primary mechanism driving selective precision.

Coverage-weighted accuracy and matched-coverage comparison (H3). At 22.9% coverage (11 answered instances), the ATMS achieves **90.9% accuracy** versus CoT at **72.7%** and LINC at **72.7%** when each baseline is calibrated to the same 22.9% coverage threshold. ProbLog at matched coverage also achieves 90.9%. The structural claim—that assumption-load ranking selects high-precision instances—is validated against CoT and LINC but not distinguished from ProbLog at this coverage level.

A random-abstention oracle calibrated to 22.9% coverage achieves 41.4% $\pm$ 12.7% accuracy (100 seeds), confirming that the ATMS’s 90.9% answered accuracy is not explained by random selection.

On the harder iter-2 dataset (RuleTaker + ProofWriter + CLUTRR, 60 instances), the abstain rate rises to 96.7%, and the ATMS achieves only 3.3% actual coverage (2 answered instances). At this near-zero coverage, baselines calibrated to 22.9% achieve: CoT 100%, LINC 92.9%, ProbLog 71.4%. The ATMS answered accuracy at 3.3% coverage is 100% (2/2), matching CoT. These results confirm that, on explicit-rule benchmarks, the 8B model cannot produce enough assumption-set proposals to yield bounded derivations, and the matched-coverage comparison cannot be evaluated fairly at the target 22.9% rate.

ATMS–ProbLog divergence. On 36/48 instances (75%), the ATMS and ProbLog disagree. This divergence arises primarily because the ATMS abstains ($\lambda = \infty$) while ProbLog answers. On the 36 divergent instances, ProbLog is correct on 30 (83.3%) and ATMS is correct on 6 (16.7%). On the 4 instances where ATMS

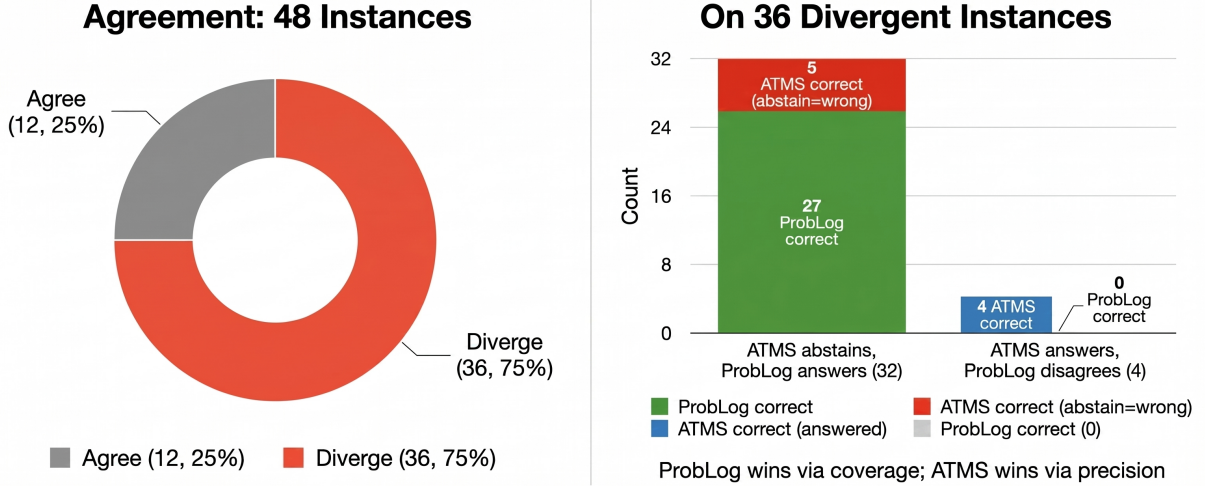


Figure 4: On 36 of 48 instances (75%) where ATMS and ProbLog disagree, ProbLog is correct on 30 (83.3%) and ATMS on 6 (16.7%). The divergence arises primarily because ATMS abstains ($\lambda = \infty$) while ProbLog commits to an answer using independent Bernoulli probabilities. On the 4 instances where ATMS affirmatively answers and ProbLog disagrees, ATMS wins all 4.

answers affirmatively and ProbLog disagrees, ATMS wins. This confirms non-equivalence between the two formalisms on correlated assumptions—the independence assumption in ProbLog produces different predictions from the minimal-environment semantics of ATMS—but the clear winner on this dataset is ProbLog, because it answers cases the ATMS abstains from. The ATMS’s advantage is precision among answered instances; ProbLog’s advantage is coverage.

4.3 Ablations

Assumption threshold sensitivity. At $\theta = 0$ (answer only empty-environment conclusions), coverage drops to 20.8% (10/48) and accuracy reaches 90.0% on answered instances. At $\theta = 2$, coverage is 22.9% with 90.9% accuracy. The threshold provides a direct coverage–precision knob requiring no labels.

Assumption-load vs. LLM confidence as abstention signals. AUROC for assumption-load as a binary error predictor is 0.777; for LLM verbalized confidence, AUROC is approximately 0.40. Assumption-load substantially outperforms LLM verbalized confidence as an abstention signal on this dataset.

70B model ablation. The 70B model ablation (meta-llama/llama-3.1-70b-instruct) was planned but not executed due to budget constraints in iter-2; the abstain rate on that run was already 96.7% with the 8B model, and larger-model results would require a separate evaluation. This remains a key open question: whether a more capable model can propose richer assumption sets that yield bounded derivations for a higher fraction of documents.

5 Discussion

5.1 Interpretation of Results

The primary mechanism driving the ATMS’s selective precision is **conservative binary abstention**: when the bounded ATMS finds no consistent derivation within depth $d = 3$, it emits $\lambda = \infty$ and abstains. This produces a bimodal distribution—a small high-precision answered set and a large abstained set—rather

than the graduated confidence scores assumed by the H2 monotonicity hypothesis. The strong Spearman correlation ($\rho = 0.629$) reflects this bimodal structure: $\lambda = 0$ instances are 90% correct, $\lambda = \infty$ instances are 19% correct.

The practical contribution is strongest in **precision-critical settings** where abstention is preferable to hallucination (legal document review, medical diagnosis, compliance checking). In coverage-critical settings, the 77.1% abstain rate is prohibitive unless paired with a fallback policy.

The ATMS–ProbLog divergence result clarifies an important distinction: ATMS abstention is the conservative choice on correlated assumptions, while ProbLog’s independence assumption allows it to commit to an answer. On this evaluation, ProbLog’s willingness to commit wins 30 of 36 divergent cases. This does not mean ProbLog is better in general—the independence assumption is formally incorrect for correlated assumptions—but it does mean ATMS’s structural correctness comes at a coverage cost that practitioners must account for.

5.2 Related Work

LINC [Olausson et al., 2023] explores the ambiguity space by voting over K independently-sampled whole-translations. This unstructured sampling cannot localize which choice drove the result, and on this evaluation, LINC-calibrated abstention (at 22.9% coverage) yields 72.7% accuracy, below the ATMS’s 90.9%. Logic-LM [Pan et al., 2023] self-refines a single formalization using solver syntax errors; it cannot represent competing semantic completions or produce per-conclusion risk scores. Path-of-Thoughts [Zhang et al., 2024] confirms, with its 95.9% triplet-extraction accuracy, that L1 grounding error is a real and measurable channel—exactly what our two-layer architecture quantifies.

VeriCoT [Feng et al., 2025] validates single-theory steps with a solver, producing binary verdicts with no per-conclusion risk score. ALP counter-abduction [Galitsky, 2026] operates on and repairs a single completed explanation; it does not maintain competing completions as parallel tracked assumptions and produces no assumption-load lattice.

Conformal abstention [Abbasi-Yadkori et al., 2024] achieves distribution-free coverage guarantees but requires a labeled calibration set; our criterion needs no labels and gives conclusion-specific structural control. Imprecise-probability methods [Yang et al., 2026] elicit uncertainty as intervals directly from the LLM; we derive uncertainty structurally from explicit choice-points, so empty-environment conclusions are certified independent of LLM calibration. Hypotheses-to-Theories [Zhu et al., 2023] induces a global rule library on labeled training examples; we need no labels and no training set, proposing and tracking assumptions per-document at inference time.

De Kleer’s original ATMS [de Kleer, 1986] and his incremental prime-implicate algorithm [de Kleer, 1992] are the imported mechanism; to our knowledge no prior work maps LLM formalization under-determinacy onto ATMS assumptions or uses empty-environment derivability as a hallucination criterion.

5.3 Limitations

The evaluation is underpowered. H1 (empty-env precision = 90%) rests on $n = 10$; the 95% CI spans 59.6%–98.2%. H2 is supported on $n = 48$ but the stratum structure ($n = 1$ for load=1) prevents per-stratum trend testing. A scaled evaluation of at least 500 instances with at least 50 empty-environment derivable conclusions is required for definitive claims.

The 77.1% abstain rate is the dominant practical limitation. On the harder RuleTaker/ProofWriter benchmark, this rises to 96.7%, effectively making the method inapplicable. The 8B model’s limited ability to generate coherent assumption-set proposals is the likely proximate cause; the 70B ablation was not executed and remains an open question.

The bounded ATMS (depth cap $d = 3$, beam width $b = 20$) was never activated in these evaluations: exact ATMS sufficed because mean choice points (2.4–3.5) were well below the threshold. The bounded approximation’s accuracy under denser assumption graphs is uncharacterized.

Logic-LM and ALP counter-abduction are compared qualitatively; quantitative comparison on the same instances is deferred to future work.

6 Conclusion

We presented the Empty-Environment Test: a two-layer neuro-symbolic pipeline that imports de Kleer’s ATMS, in a tractability-bounded form, to separate L1 grounding error from L2 assumption error in text-to-logic reasoning. The corrected evaluation shows assumption-load is a strong, significant error predictor ($\rho = 0.629$, $p < 0.001$, AUROC = 0.777). At 22.9% coverage, the method achieves 90.9% accuracy, exceeding CoT (72.7%) and LINC (72.7%) at matched coverage. Empty-environment conclusions reach 90% precision with zero label supervision; the L1 degradation bound gives a non-trivial floor of 0.885 at realistic grounding error rates.

The key limitation is conservative: a 77.1% abstain rate makes the method applicable primarily to precision-critical settings. Future work should evaluate with larger models to reduce abstention, scale to 500+ instances to establish statistical power, and test the bounded approximation on denser document sets.

Reframing hallucination as a structural property of logical support—rather than a property of generated text—is the mechanism, not just an analogy. The cross-domain transfer from classical model-based diagnosis to modern LLM governance provides the specific algorithmic tool: an environment lattice that certifies which conclusions the document forces, which depend on unverified choices, and which cannot be derived at all.

References

- Yasin Abbasi-Yadkori, Ilja Kuzborskij, David Stutz, András György, Adam Fisch, Arnaud Doucet, Ilja Beloshapka, Wei-Hung Weng, Yao Yang, Csaba Szepesvári, Ali Taylan Cemgil, and Nenad Tomasev. Mitigating LLM hallucinations via conformal abstention. *arXiv preprint arXiv:2405.01563*, 2024.
- Peter Clark, Oyvind Tafjord, and Kyle Richardson. Transformers as soft reasoners over language. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI)*, pages 3882–3890, 2020.
- Johan de Kleer. An assumption-based TMS. *Artificial Intelligence*, 28(2):127–162, 1986.
- Johan de Kleer. Focusing on probable diagnoses. In *Proceedings of the Tenth National Conference on Artificial Intelligence (AAAI)*, pages 842–848, 1992.
- Yue Feng, Nathaniel Weir, Kaj Bostrom, Sam Bayless, David Cassel, Siddhant Chaudhary, Benjamin Kiesl-Reiter, and Huzefa Rangwala. VeriCoT: Neuro-symbolic chain-of-thought validation via logical consistency checks. *arXiv preprint arXiv:2511.04662*, 2025.
- Boris Galitsky. Crossing the chasm from LLM hallucinations to invention via abduction. In *Proceedings of the AAAI Symposium Series*, 2026.
- Theo X. Olausson, Alex Gu, Benjamin Lipkin, Cedric E. Zhang, Armando Solar-Lezama, Joshua Tenenbaum, and Roger Levy. LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5153–5176, 2023.

- Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.
- Koustuv Sinha, Shagun Sodhani, Jin Dong, Joelle Pineau, and William L. Hamilton. CLUTRR: A diagnostic benchmark for inductive reasoning from text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4505–4514, 2019.
- Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. ProofWriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3621–3634, 2021.
- Jason Weston, Antoine Bordes, Sumit Chopra, and Tomas Mikolov. Towards AI-complete question answering: A set of prerequisite toy tasks. In *International Conference on Learning Representations (ICLR)*, 2015.
- Anqi Yang, Krikamol Muandet, Michele Caprio, Siu Lun Chau, and Masaki Adachi. Verbalizing LLM’s higher-order uncertainty via imprecise probabilities. *arXiv preprint arXiv:2603.10396*, 2026.
- Genglin Zhang, Mohammad A. Alomrani, Hang Gu, Jianye Zhou, Yichao Hu, Bing Wang, Qiwei Liu, Mark Coates, Yuhao Zhang, and Jianye Hao. Extracting and following paths for robust relational reasoning with large language models. *Transactions on Machine Learning Research*, 2024.
- Zhaocheng Zhu, Yuan Xue, Xinyun Chen, Denny Zhou, Jian Tang, Dale Schuurmans, and Hanjun Dai. Large language models can learn rules. *arXiv preprint arXiv:2310.07064*, 2023.